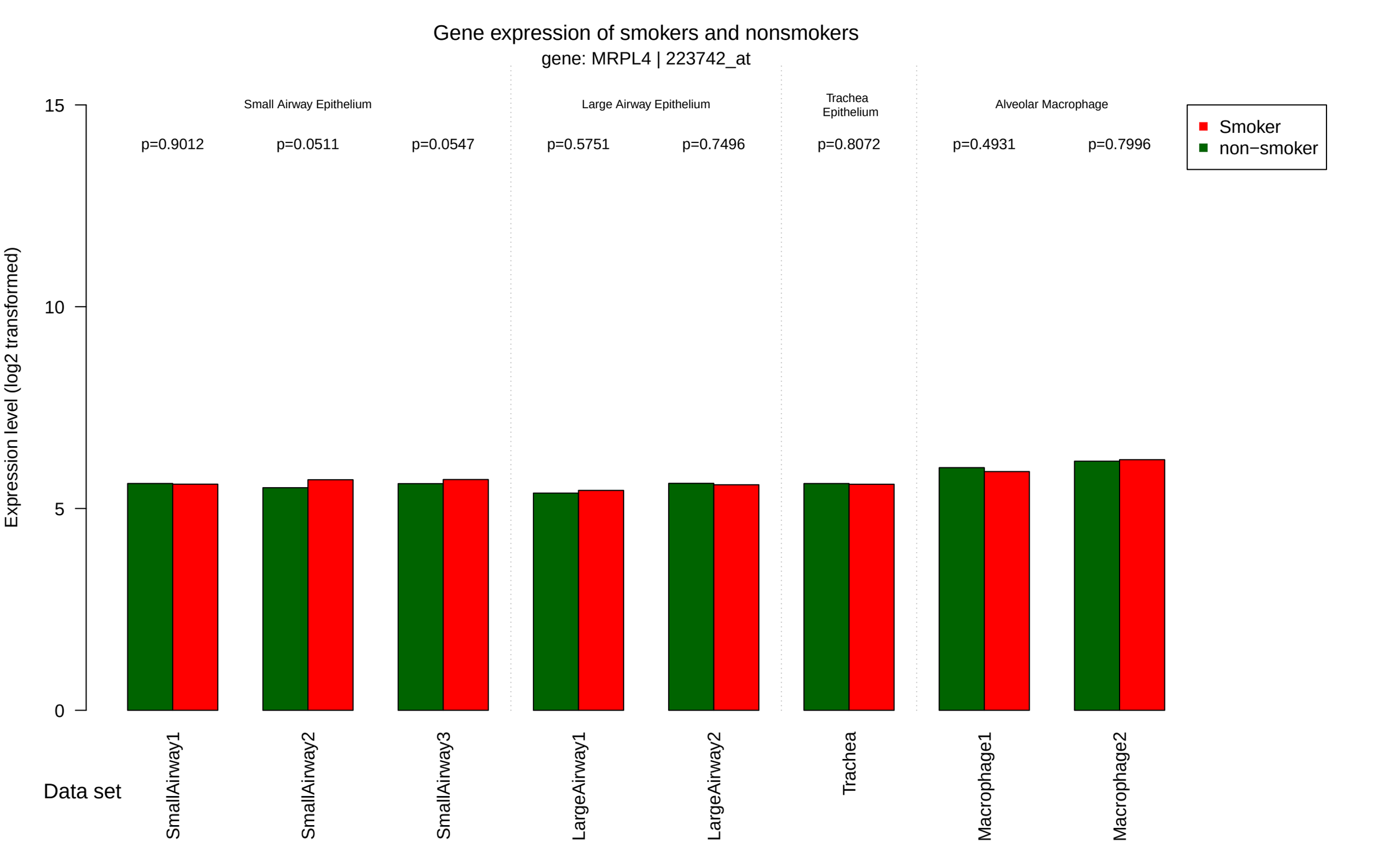
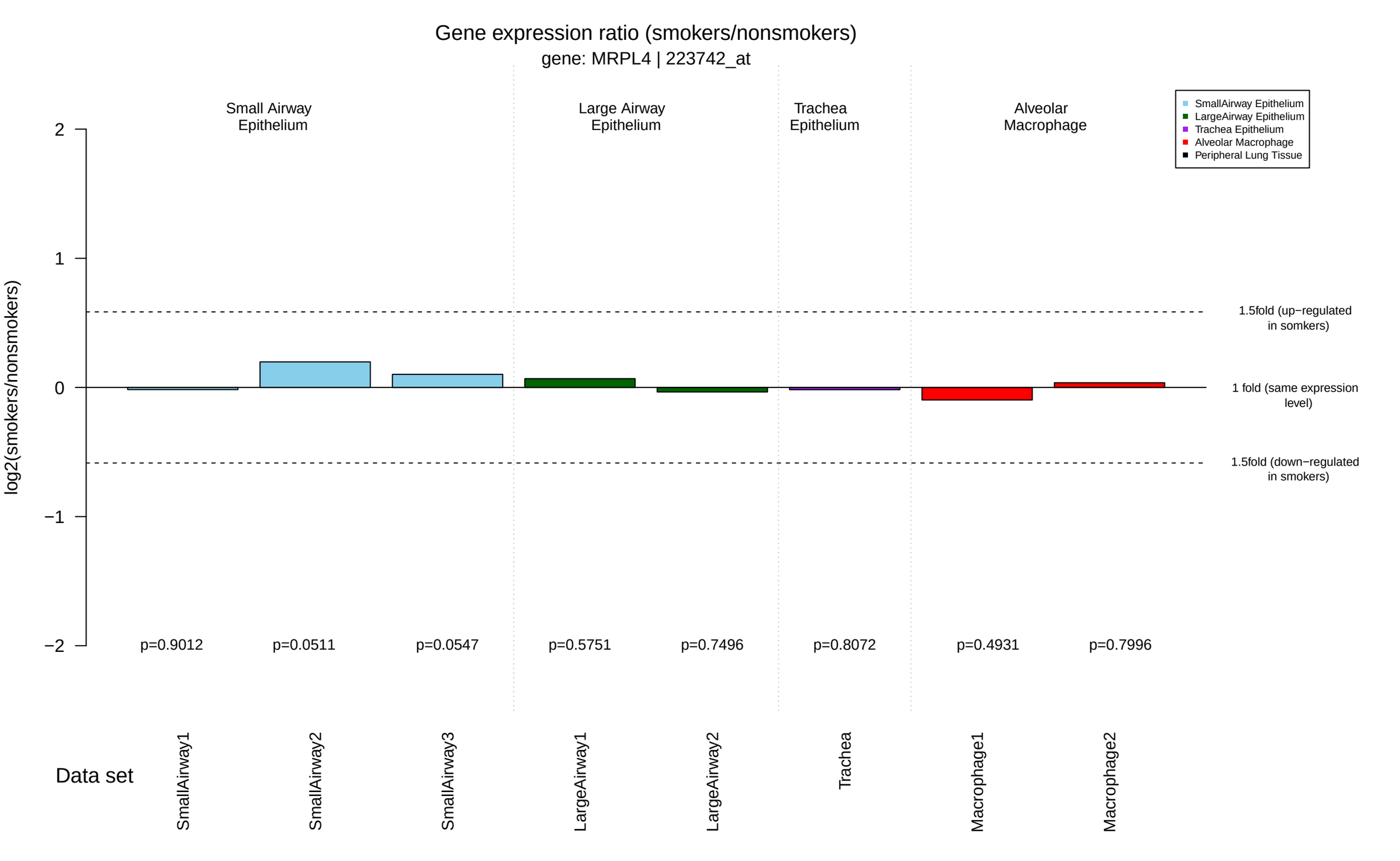
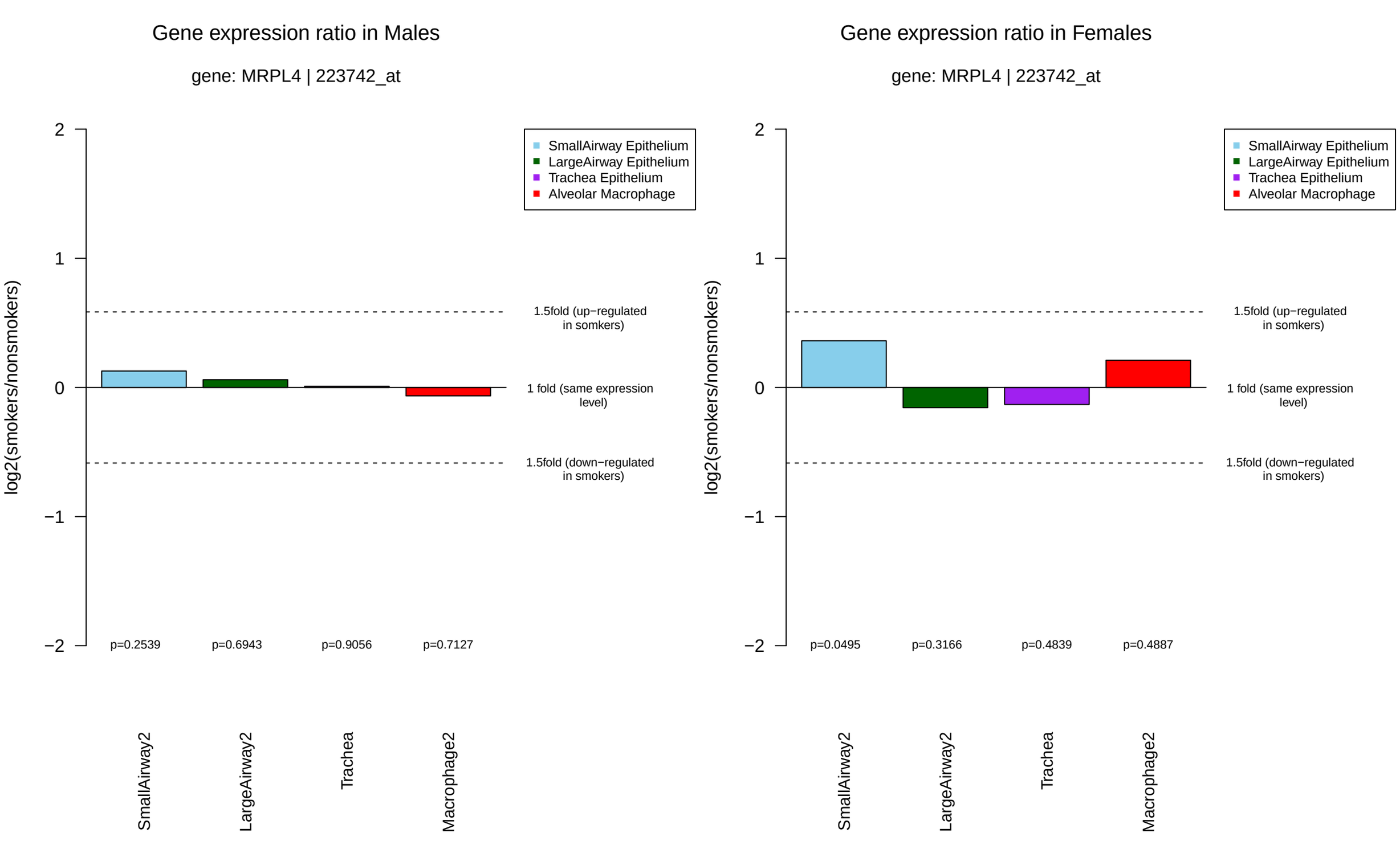
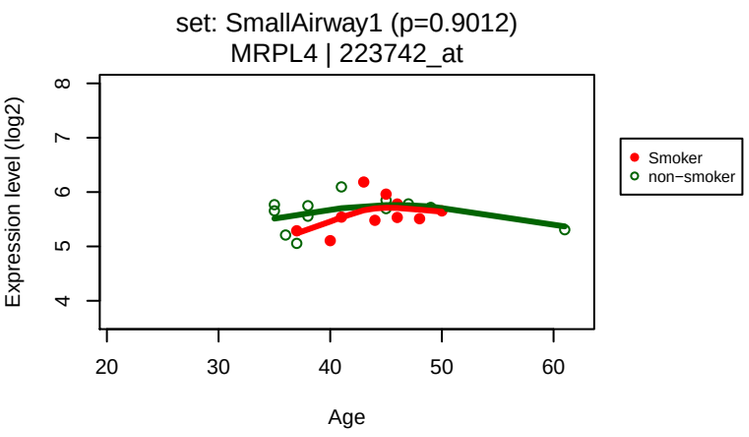
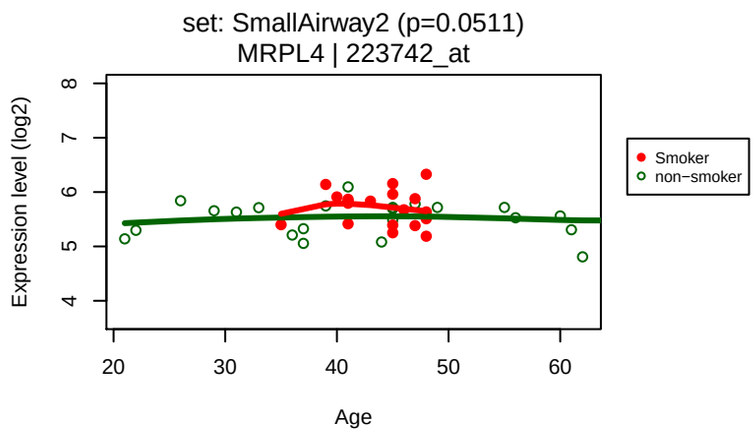
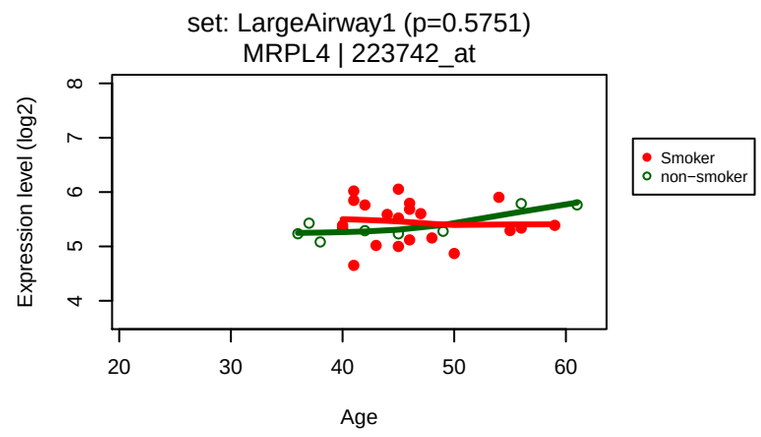

Gene expression of smokers and nonsmokers
gene: MRPL4 | 223742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9012
p=0.0511
p=0.0547
p=0.5751
p=0.7496
p=0.8072
p=0.4931
p=0.7996
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

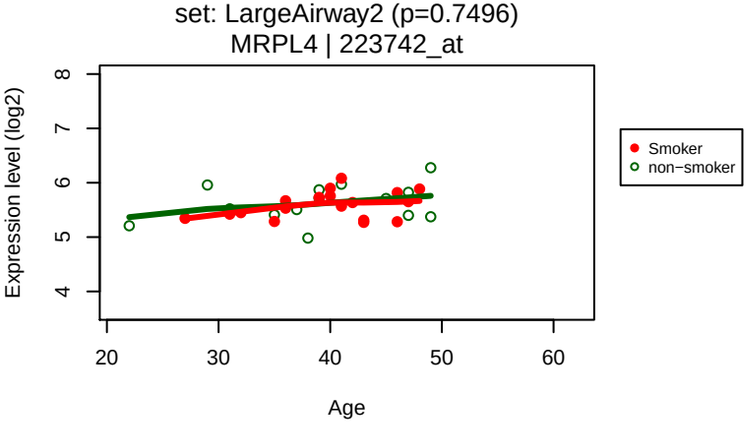

Gene expression ratio (smokers/nonsmokers)
gene: MRPL4 | 223742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9012
p=0.0511
p=0.0547
p=0.5751
p=0.7496
p=0.8072
p=0.4931
p=0.7996
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

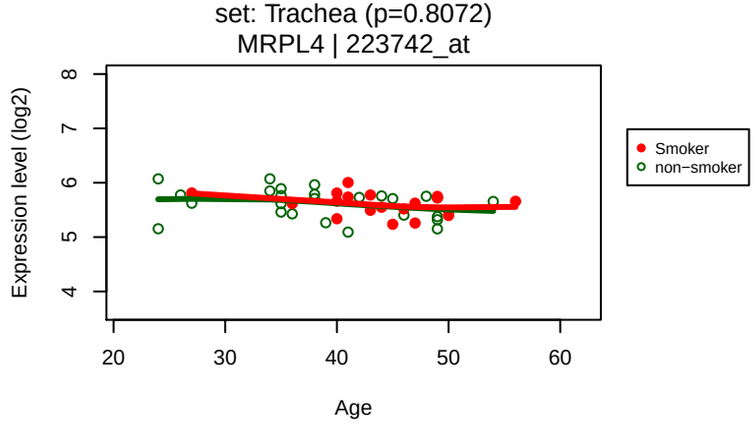

Gene expression ratio in Males
gene: MRPL4 | 223742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.6943
p=0.9056
p=0.7127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL4 | 223742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0495
p=0.3166
p=0.4839
p=0.4887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

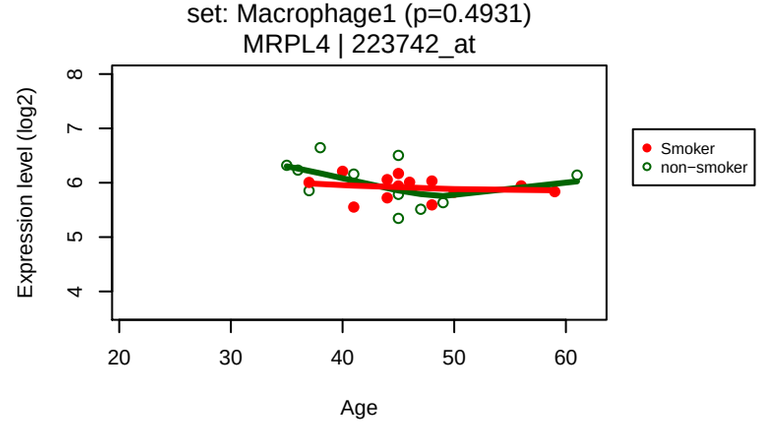

set: SmallAirway1 (p=0.9012)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker

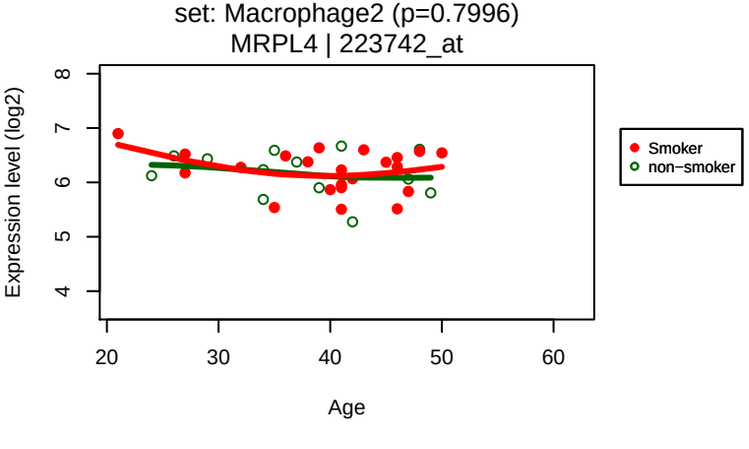

set: SmallAirway2 (p=0.0511)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker

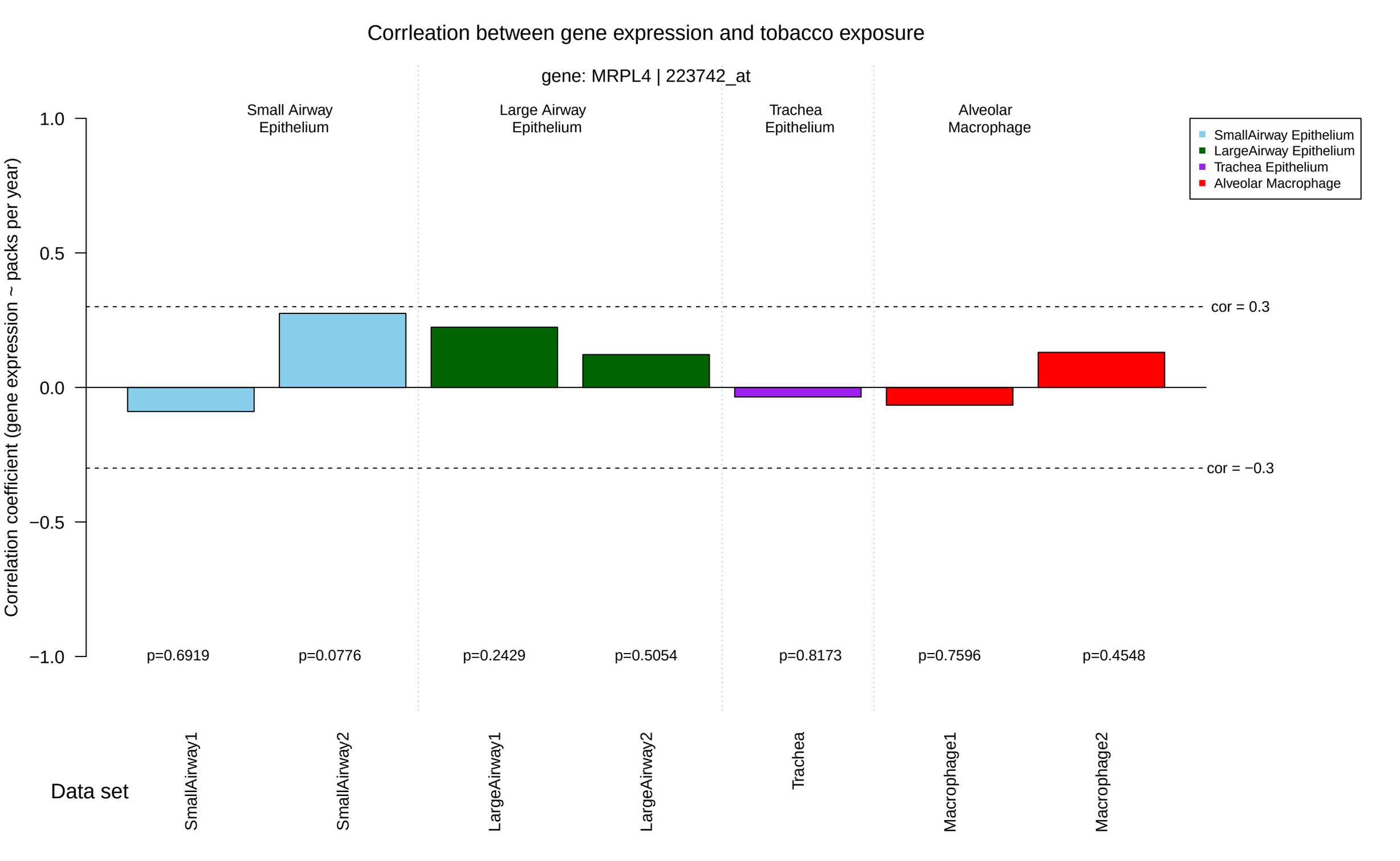

set: LargeAirway1 (p=0.5751)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7496)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8072)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4931)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7996)
MRPL4 | 223742_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL4 | 223742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6919
p=0.0776
p=0.2429
p=0.5054
p=0.8173
p=0.7596
p=0.4548
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2